

78 - Creating roles

RoleManager in ASP.NET Core

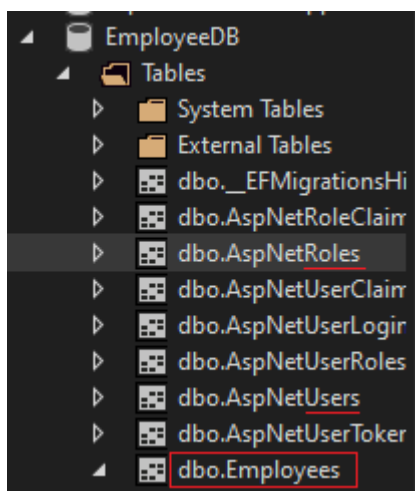
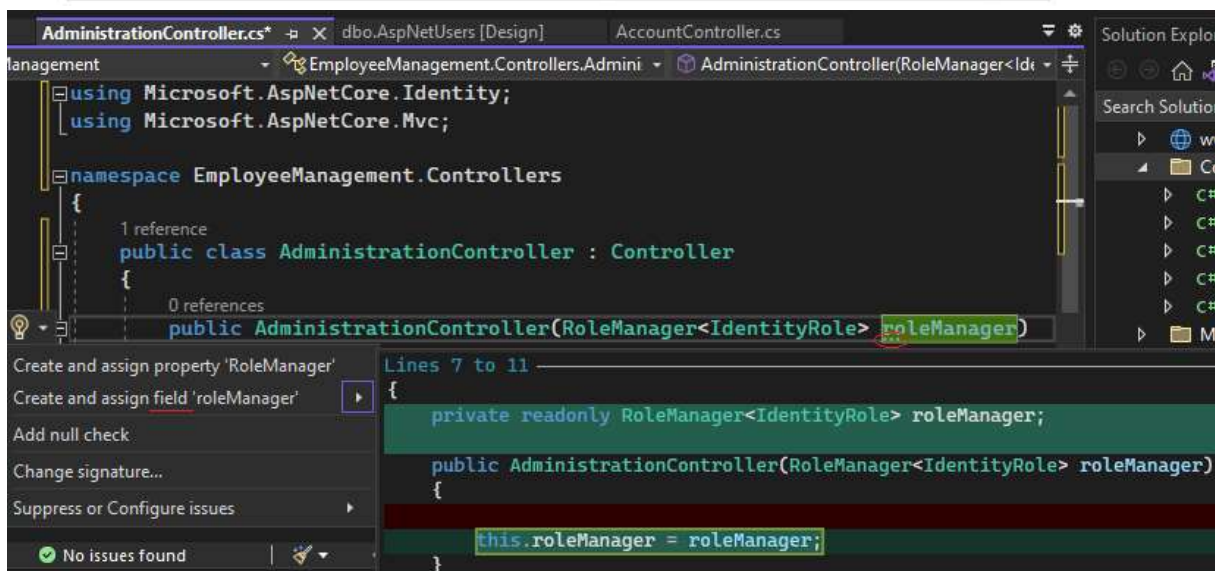
UserManager<IdentityUser>

RoleManager<IdentityRole>

```
public class AdministrationController : Controller
{
    private readonly RoleManager<IdentityRole> roleManager;

    public AdministrationController(RoleManager<IdentityRole> roleManager)
    {
        this.roleManager = roleManager;
    }

    // Rest of the code
}
```



The screenshot shows the 'dbo.AspNetRoles' table in the 'Data' view. The table has columns: Id, Name, NormalizedName, and ConcurrencyStamp. The data is as follows:

Id	Name	NormalizedName	ConcurrencyStamp
NULL	NULL	NULL	NULL

Като изключим таблица Employees, останалите са генерирани от Identity API (app programming

interface). Към момента нямаме попълнени Roles. За да го създадем, ни трябва само име, а ID ще бъде автоматично попълнено, като за целта трябва да създадем ViewModel.

```

CreateRoleViewModel.cs
using System.ComponentModel.DataAnnotations;

namespace EmployeeManagement.ViewModels
{
    3 references
    public class CreateRoleViewModel
    {
        [Required]
        3 references
        public string RoleName { get; set; }
    }
}

```

```

AdministrationController.cs*  X  dbo.AspNetUsers [Design]
EmployeeManagement.Controllers.Admini  CreateRole()

using Microsoft.AspNetCore.Identity;
using Microsoft.AspNetCore.Mvc;

namespace EmployeeManagement.Controllers
{
    1 reference
    public class AdministrationController : Controller
    {
        private readonly RoleManager<IdentityRole> roleManager;

        0 references
        public AdministrationController(RoleManager<IdentityRole> roleManager)
        {
            this.roleManager = roleManager;
        }

        [HttpGet]
        0 references
        public IActionResult CreateRole()
        {
            return View();
        }
    }
}

```

```
using Microsoft.AspNetCore.Identity;
using Microsoft.AspNetCore.Mvc;

namespace EmployeeManagement.Controllers
{
    public class AdministrationController : Controller
    {
        private readonly RoleManager<IdentityRole> roleManager;

        public AdministrationController(RoleManager<IdentityRole> roleManager)
        {

```

```

        this.roleManager = roleManager;
    }
    [HttpGet]
    public IActionResult CreateRole()
    {
        return View();
    }
}

```

Views->Account->CreateRole.cshtml, copy->paste from Register.cshtml + modify:

```

CreateRole.cshtml*  CreateRoleViewModel.cs  AdministrationController.cs
@model CreateRoleViewModel
@{
    ViewBag.Title = "Create New Role";
}

<h1>Create New Role</h1>

<div class="row">
    <div class="col-md-12">
        <form method="post">
            <div asp-validation-summary="All" class="text-danger"></div>
            <div class="form-group">
                <label asp-for="RoleName"></label>
                <input asp-for="RoleName" class="form-control" />
                <span asp-validation-for="RoleName" class="text-danger"></span>
            </div>
            <button style="width:auto" type="submit"
                class="btn btn-primary">Create Role</button>
        </form>
    </div>
</div>

```

```

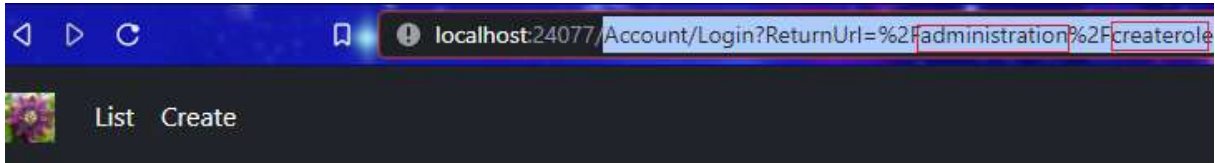
@model CreateRoleViewModel
@{
    ViewBag.Title = "Create New Role";
}

<h1>Create New Role</h1>

<div class="row">
    <div class="col-md-12">
        <form method="post">
            <div asp-validation-summary="All" class="text-danger"></div>
            <div class="form-group">
                <label asp-for="RoleName"></label>
                <input asp-for="RoleName" class="form-control" />
                <span asp-validation-for="RoleName" class="text-danger"></span>
            </div>
            <button style="width:auto" type="submit"
                class="btn btn-primary">Create Role</button>
        </form>
    </div>
</div>

```

Трябва да сме LogOut, за да проверим: /administration/createrole (controller/action)



User Login

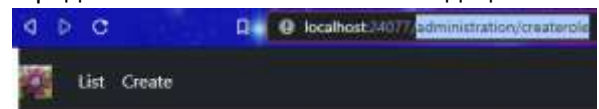
Email

Password

☐ Remember me

Login

Заради global authorizeFilter, изисква регистрация преди CreateNewRole. Има и валидация



```
Startup.cs | CreateRoleViewModel.cs | AdministrationCo
- EmployeeManagement.Startup | ConfigureServices(Se
services.AddMvc(options =>
{
    options.EnableEndpointRouting = false;
    var policy = new AuthorizationPolicyBuilder()
        .RequireAuthenticatedUser()
        .Build();
    options.Filters.Add(new AuthorizeFilter(policy));
}).AddXmlSerializerFormatters();
```

Create New Role

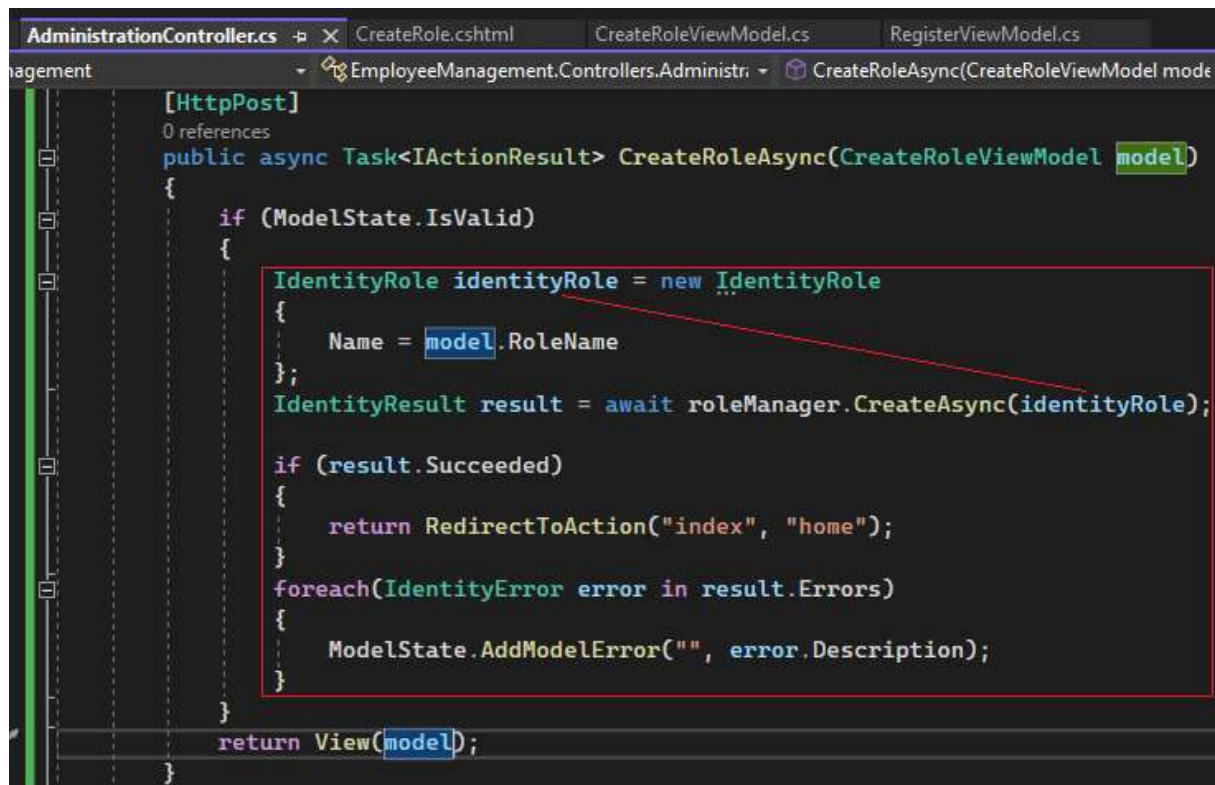
• The RoleName field is required.

RoleName

The RoleName field is required.

Create Role

Ако се опитаме да въведем role – Admin, понеже все още нямаме Post -> 404.



```
[HttpPost]
0 references
public async Task<IActionResult> CreateRoleAsync(CreateRoleViewModel model)
{
    if (ModelState.IsValid)
    {
        IdentityRole identityRole = new IdentityRole
        {
            Name = model.RoleName
        };
        IdentityResult result = await roleManager.CreateAsync(identityRole);

        if (result.Succeeded)
        {
            return RedirectToAction("index", "home");
        }
        foreach (IdentityError error in result.Errors)
        {
            ModelState.AddModelError("", error.Description);
        }
    }

    return View(model);
}
```

```
using EmployeeManagement.ViewModels;
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Identity;
using Microsoft.AspNetCore.Mvc;
using System.Threading.Tasks;

namespace EmployeeManagement.Controllers
{
    public class AdministrationController : Controller
    {
        private readonly RoleManager<IdentityRole> roleManager;

        public AdministrationController(RoleManager<IdentityRole> roleManager)
        {
            this.roleManager = roleManager;
        }

        [HttpGet]
        public IActionResult CreateRole()
        {
            return View();
        }

        [HttpPost]
        public async Task<IActionResult> CreateRoleAsync(CreateRoleViewModel
model)
        {
            if (ModelState.IsValid)
            {
                IdentityRole identityRole = new IdentityRole
                {
                    Name = model.RoleName
                };
                IdentityResult result = await
roleManager.CreateAsync(identityRole);
```



```

        if (result.Succeeded)
        {
            return RedirectToAction("index", "home");
        }
        foreach (IdentityError error in result.Errors)
        {
            ModelState.AddModelError("", error.Description);
        }
    }
    return View(model);
}
}
}

```

Създаваме Role -> Admin, прехвърля ни в index, проверяваме в DB:

Startup.cs dbo.AspNetRoles [Data] AdministrationController.cs				
Max Rows: 1000				
	Id	Name	NormalizedNa...	ConcurrencySt...
▶	139-ebbb0d1244ae	Admin	ADMIN	05fa09d0-0f8c-...
⚙	NULL	NULL	NULL	NULL

Creating Roles in ASP.NET Core

```

[HttpPost]
public async Task<IActionResult> CreateRole(CreateRoleViewModel model)
{
    if (ModelState.IsValid)
    {
        IdentityRole identityRole = new IdentityRole { Name = model.RoleName };
        IdentityResult result = await roleManager.CreateAsync(identityRole);

        if (result.Succeeded)
        {
            return RedirectToAction("index", "home");
        }

        foreach (IdentityError error in result.Errors)
        {
            ModelState.AddModelError("", error.Description);
        }
    }

    return View(model);
}

```